

Santhosh Sanka

+91 9440425036 | santhoshsanka100@gmail.com | [linkedin.com/in/santhosh-sanka](https://www.linkedin.com/in/santhosh-sanka) | github.com/Santhosh100sks

Personal Summary

Aspiring Software Engineer pursuing B.Tech in Computer Science (Artificial Intelligence) at Amrita Vishwa Vidyapeetham with strong foundations in Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems, and Full-Stack Web Development. Experienced in building scalable web applications and AI-powered software solutions using Python, Java, JavaScript, React, Node.js, SQL, and Git. Passionate about solving real-world engineering problems through clean, efficient, and maintainable software.

Education

Amrita Vishwa Vidyapeetham, Bengaluru <i>B.Tech – Computer Science and Engineering (AI), CGPA: 8.41</i>	2024 – Present
Sri Chaitanya Junior College, Vijayawada <i>Intermediate (Class XII), Grade: 95.3%</i>	2022 – 2024
Sri Chaitanya Schools, Rajahmundry <i>SSC 10th Grade: 90.66%</i>	2021 – 2022

Technical Skills

Languages: Python, Java, C
Frontend: HTML5, CSS3, JavaScript, React.js
Backend: Node.js, Express.js
Database: MongoDB, SQL
ML / AI: Scikit-learn, XGBoost, TensorFlow, ProtBERT, LLMs, Prompt Engineering
Core CS: DSA, OOP, DBMS
Tools: Git, GitHub, Streamlit, Solara, Mesa ABM

Soft Skills

Problem Solving, Team Collaboration, Analytical Thinking, Communication, Adaptability

Projects

Respiratory Sound Classification <i>Healthcare AI</i> GitHub	May 2026
- Developed a hybrid ML/DL pipeline on the ICBHI Respiratory Sound Database for automated classification of wheezes, crackles, and mixed respiratory anomalies. - Extracted MFCC features and trained machine learning classifiers, achieving 98.8% Level-1 accuracy; fine-tuned MobileNetV2 on Mel-spectrograms, achieving 96.63% accuracy.	
Intelligent Emergency Recommendation Platform <i>Real-Time Emergency Recommendation System</i> GitHub	Oct 2025
- Built an offline agent-based emergency recommendation system using Mesa, modeling six autonomous service categories with a greedy Euclidean nearest-neighbor algorithm, achieving 94.6% top-1 accuracy and sub-125 ms response time. - Developed a Solara-based interactive web application with geospatial maps, one-tap emergency calling, and a JSON-based live center registration system.	
AF Gene Sequence Classifier <i>Machine Learning</i> <i>Bioinformatics</i> GitHub	Jul 2026
- Developed an end-to-end machine learning pipeline to classify 10,544 protein sequences into 16 Atrial Fibrillation-associated genes using ProtBERT-BFD embeddings and handcrafted physicochemical features. - Engineered a 1,051-dimensional feature space and evaluated XGBoost, Random Forest, KNN, and Voting Ensemble models, with XGBoost achieving 98.4% classification accuracy. - Optimized preprocessing using k-mer cosine similarity, achieving approximately 900× faster sequence comparison than global alignment, and deployed the solution as an interactive Streamlit web application.	

Links

LinkedIn: [linkedin.com/in/santhosh-sanka](https://www.linkedin.com/in/santhosh-sanka) GitHub: github.com/Santhosh100sks

Languages

English, Telugu, Hindi (Basic)

Certifications

Generative AI in Action — IBM
Machine Learning Operations (MLOps) for Generative AI – Google Skills